KnowSelf

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

arxiv.org/abs/2504.03553

Current Agent Training Uses 'Flood Irrigation'

Indiscriminate Knowledge Injection Hurts Efficiency

Direct Trajectory Imitation

ReAct, etc.
Copies gold trajectories
at every step regardless
of need.

Limitation: Agent never learns
when it could act on its own

Trial-and-Error Refinement

Reflexion, ETO, etc.
Injects external feedback
after every failed step,
even when unnecessary.

Limitation: Wasteful reflection
on simple steps

Knowledge-Augmented Planning

KnowAgent, WKM, ExpeL
Applies domain knowledge
at every step (100%),
ignoring self-sufficiency.

Limitation: Fragile plans, high
inference cost, pattern collapse

Humans possess situational self-awareness — the metacognitive ability to assess whether they can handle a situation directly, need to pause and reflect, or must seek external help.

KnowSelf brings this capability to LLM agents: instead of flood-irrigating knowledge at every step, the agent learns to autonomously decide when to act, reflect, or consult external knowledge.

Three Situational Thinking Modes

KnowSelf classifies each decision step based on the agent's capability at that moment

Fast Thinking [Direct]

Predicted action matches gold action

Agent can directly produce correct action

No reflection or external knowledge needed

Token: <Direct>

Slow Thinking [Reflect]

Initial action wrong, but self-reflection corrects it

Agent initially produces an incorrect action

Reflection enables self-correction

Token: <Reflection>

Knowledgeable Thinking [Knowledge]

Cannot correct even after reflection

Agent cannot reach correct action via reflection

Must consult external knowledge base

Token: <Knowledge>

KnowSelf Framework

Data Construction → Two-Stage Training → Self-Aware Inference

1 Stage 1: Self-Awareness Data Construction

1. Agent self-explores the environment to generate trajectories

2. Heuristic criterion classifies each step:

- Action = Gold → Fast (Direct)
- Reflection fixes → Slow (Reflect)
- Otherwise → Knowledgeable

3. Special tokens inserted to mark each situation type

Key: Lightweight heuristic labeling on self-explored data — no expensive human annotation

2 Stage 2: Two-Stage Training

Phase A: Supervised Fine-Tuning (SFT)

Teaches initial self-awareness patterns

Agent learns to predict special tokens

Phase B: Reward-based Policy Optimization (RPO)

Refines token generation ability

Reinforces correct situational decisions

Improves calibration of when to seek help

3 Stage 3: Self-Aware Inference

At deployment: Agent generates special tokens autonomously

<Direct> → Proceed with action immediately

No extra computation needed

<Reflection> → Pause and self-reflect

Agent reconsiders before acting

<Knowledge> → Query external KB

Retrieve relevant domain knowledge

Qualitative Examples: KnowSelf vs. Strong Baselines

KnowSelf correctly heats a mug and places a saltshaker where baselines fail

"Put a hot mug in cabinet"

KnowSelf uses: Reflection Token

OpenAI-O1 (fails)

Opens the microwave and incorrectly tries to take an egg from it — wrong object entirely.

KnowSelf (Llama-8B, succeeds)

Generates <Reflection> token → realizes it should heat the mug using the open microwave. Self-corrects before acting on wrong object.

✦ Reflection enables self-correction without external knowledge

"Put a saltshaker in drawer"

KnowSelf uses: Knowledge Token

DeepSeek-R1 (fails)

Reflects that the drawer is open and tries to close it first — unnecessary action that fails.

KnowSelf (Llama-8B, succeeds)

Generates <Knowledge> token → retrieves rule: place the object without removing unrelated items. Directly places saltshaker in the open drawer.

✦ External knowledge prevents common-sense errors that reflection alone cannot fix

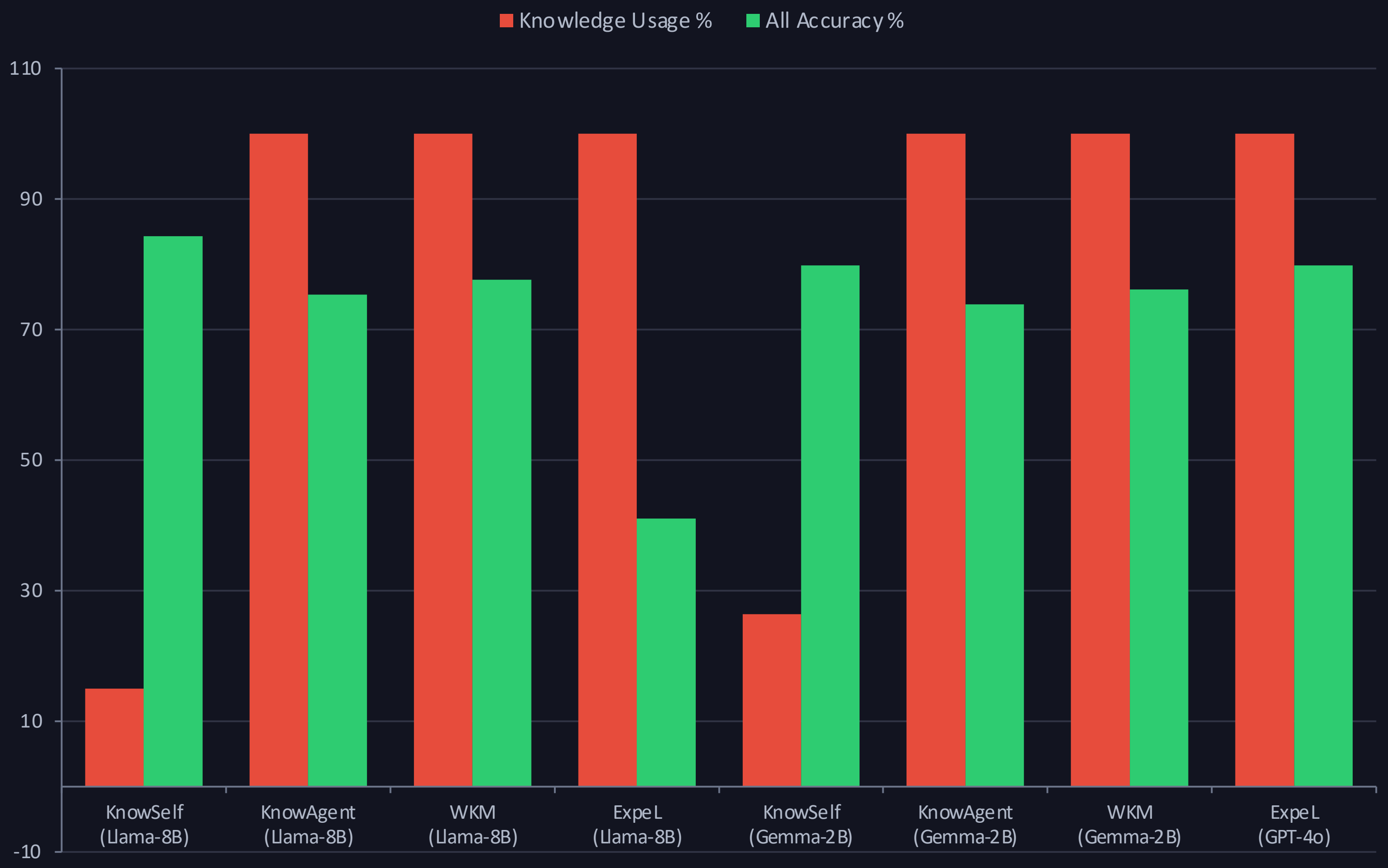
Main Results on ALFWorld

KnowSelf achieves 84.33% with Llama-8B using only 15% knowledge — surpassing GPT-4o ExpeL (79.85%) at 100% knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	**KnowSelf**	**15.01%**	**91.67**	**87.10**	**91.30**	**85.71**	**77.78**	**64.71**	**84.33**
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	**KnowSelf**	**26.41%**	**87.50**	**93.55**	**73.91**	**76.19**	**83.33**	**52.94**	**79.85**

Minimal Knowledge, Maximum Impact

KnowSelf uses only 15-26% knowledge queries yet achieves the best overall scores across model scales



15% Knowledge, 84.33% Accuracy

KnowSelf (Llama-8B) uses only 15.01% external knowledge yet achieves the highest overall score among all Llama-8B methods.

100% Knowledge ≠ Better

KnowAgent, WKM, and ExpeL all use 100% knowledge but underperform KnowSelf. Indiscriminate knowledge injection hurts planning quality.

Scales Down Effectively

Even Gemma-2B with KnowSelf reaches 79.85% with only 26.41% knowledge — matching GPT-4o ExpeL that uses 100% knowledge.

Key Takeaways

Self-Awareness Is the Missing Piece in Agent Planning

1. **Indiscriminate knowledge injection is wasteful and suboptimal**

   Methods using 100% knowledge (ExpeL, KnowAgent, WKM) often underperform KnowSelf's selective approach — more knowledge is not always better.

2. **Agents can learn situational self-awareness via special tokens with lightweight heuristic labeling**

   No expensive human annotation or massive external feedback needed — only heuristic classification on self-explored trajectories.

3. **KnowSelf (Llama-8B) outperforms GPT-4o baselines: 84.33% vs. 76.12% (ReAct) and 79.85% (ExpeL)**

   A smaller open-source model with self-awareness beats larger proprietary models relying on brute-force knowledge injection.

4. **Only 15-26% knowledge usage needed versus 100% for competing methods**

   Selective knowledge retrieval dramatically reduces inference cost while improving planning quality — efficiency and effectiveness together.

5. **The paradigm generalizes across model scales — even Gemma-2B reaches 79.85%, matching GPT-4o ExpeL**

   KnowSelf's benefits are not limited to large models: the self-awareness mechanism transfers effectively to smaller architectures.

Future Directions: Extending to multi-agent settings · Transfer across task domains · Richer self-awareness taxonomies · Integration with tool-use planning